

Micah Kepe

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EXPERIENCE

Software Engineering Intern

Jan 2026 – May 2026

Atomic Semi

San Francisco, CA

- Architected a real-time collaborative chip design suite (layout and schematic editors) in Rust compiled to WebAssembly, with multiplayer state synchronization and Dioxus reactive components.
- Sustained >100 FPS interactive rendering of semiconductor layouts through optimized rendering pipelines and instance rendering; built browser and native flamegraph profiling to identify bottlenecks.
- Wrote a zero-copy streaming GDSII parser at ≈ 2 GB/s throughput; open-sourced on crates.io (github.com/micahkepe/gdsii).
- Implemented Python FFI for parametric cells (PCells) and serialization and schema infrastructure; contributed upstream to [Dioxus](https://dioxus.rs) and [Gitea](https://gitea.io).

Undergraduate Research Assistant

May 2025 – Dec 2025

Rice University

Houston, TX

- Advised by Dr. Konstantinos Mamouras; designed high-performance streaming algorithms in Rust for querying semi-structured data (e.g., JSON) using finite automata techniques.
- Achieved 86–95% faster performance over leading JSON query tools by compiling queries into optimized state machines; built throughput-optimized pipelines processing 100+ MB/s.

Machine Learning Research Intern

Feb 2024 – Aug 2024

Vislang Lab

Houston, TX

- Advised by Dr. Vicente Ordóñez-Román; co-led development of a 40M+ image repository for open-source multimodal ML research.
- Built infrastructure for dataset annotation, visualization, and semantic search (Faiss-based retrieval), enabling scalable exploration of a 100M+ image-text corpus.

PROJECTS

jsongrep | *Rust, Automata Theory, Language Design*

- 600+ GitHub stars, 3,000+ crates.io downloads; [Hacker News front page](#), featured in This Week in Rust, and listed in awesome-rust.
- Published in Homebrew-Core and winget; delivers sub-20 ms queries on 500 MB+ JSON/YAML/TOML documents via automata-based query compilation.

vimtutor-sequel | *Bash, Vimscript, Ruby, Homebrew*

- 500+ GitHub stars, [Hacker News front page](#); packaged into Homebrew-Core for use by thousands of developers worldwide.
- Authored 8 advanced Vim tutorials covering macros, registers, and plugin development.

Technical Blog | *micahkepe.com/blog*

- 20k+ Google Search impressions; writing on systems programming, developer tooling, and potpourri topics.
- Several [HackerNews front page submissions](#).

TECHNICAL SKILLS

Languages: Rust, Go, Python, C, JavaScript/TypeScript, SQL

Technologies: WebAssembly, Linux, Git, GitHub Actions, Neovim

Specializations: Systems Programming, Concurrency, Performance Optimization, Finite Automata, Developer Tooling, Cross-Language Interop

EDUCATION

Rice University

Aug 2022 – Dec 2025

Bachelor of Science in Computer Science (GPA: 3.89/4.00, cum laude)

Houston, TX